Railway Information Dataset: Data Engineering Analysis

Cognifyz Solutions Internship Task

Input: Railway_info.csv | Rows: 11,113 | Columns: 5

1. Executive Summary

Metric	Result
Rows in input	11,113
Rows after exact duplicate removal	11,113
Unique source stations	921
Unique destination stations	924
Most common source	CST-MUMBAI (513)
Most common destination	CST-MUMBAI (514)
Highest day count	Friday (1,649)
Lowest day count	Monday (1,503)

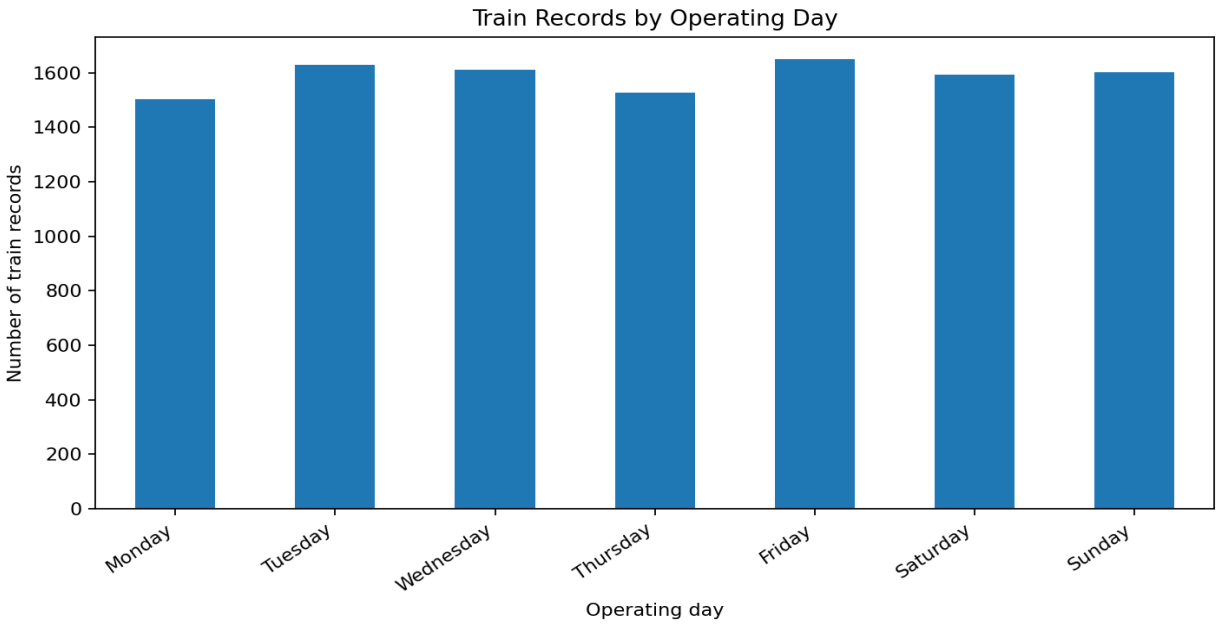
2. Data Quality and Cleaning

The dataset has columns Train_No, Train_Name, Source_Station_Name, Destination_Station_Name, and days. Missing values in the five original columns: No missing values detected in the input columns.. Exact duplicate rows: 0. Names were trimmed and standardized to uppercase. The raw days field is retained as days_raw.

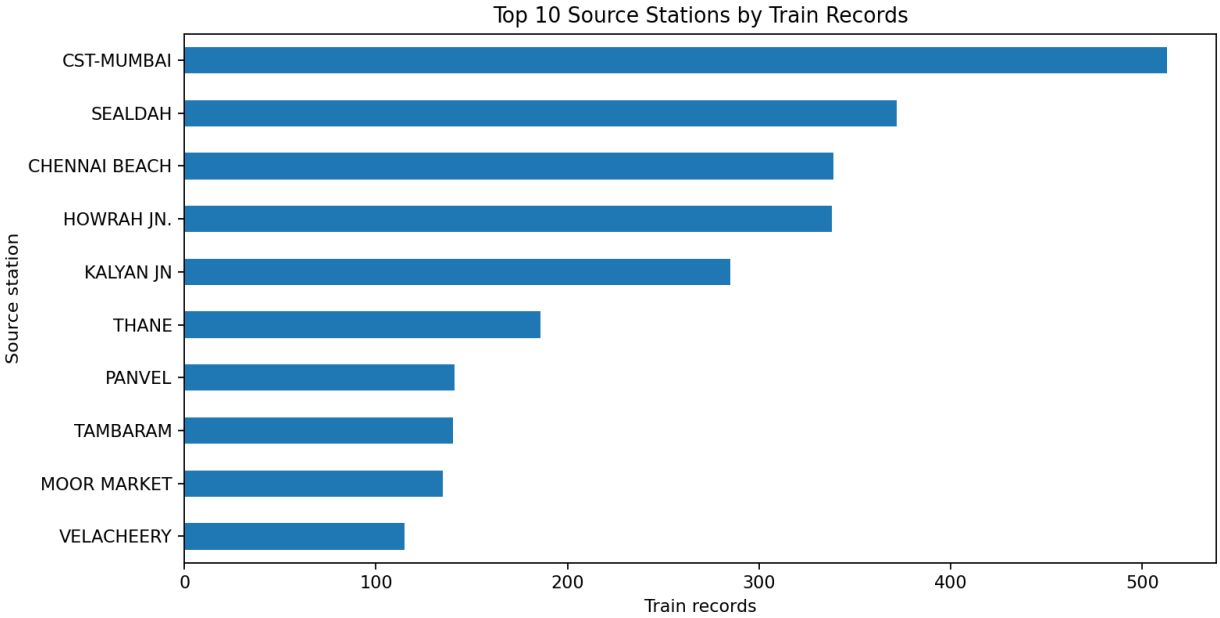
Interpretation note: values include weekday names with a trailing 'd'. The suffix is preserved and flagged; its business meaning should be confirmed with the dataset owner.

3. Analysis and Visualizations

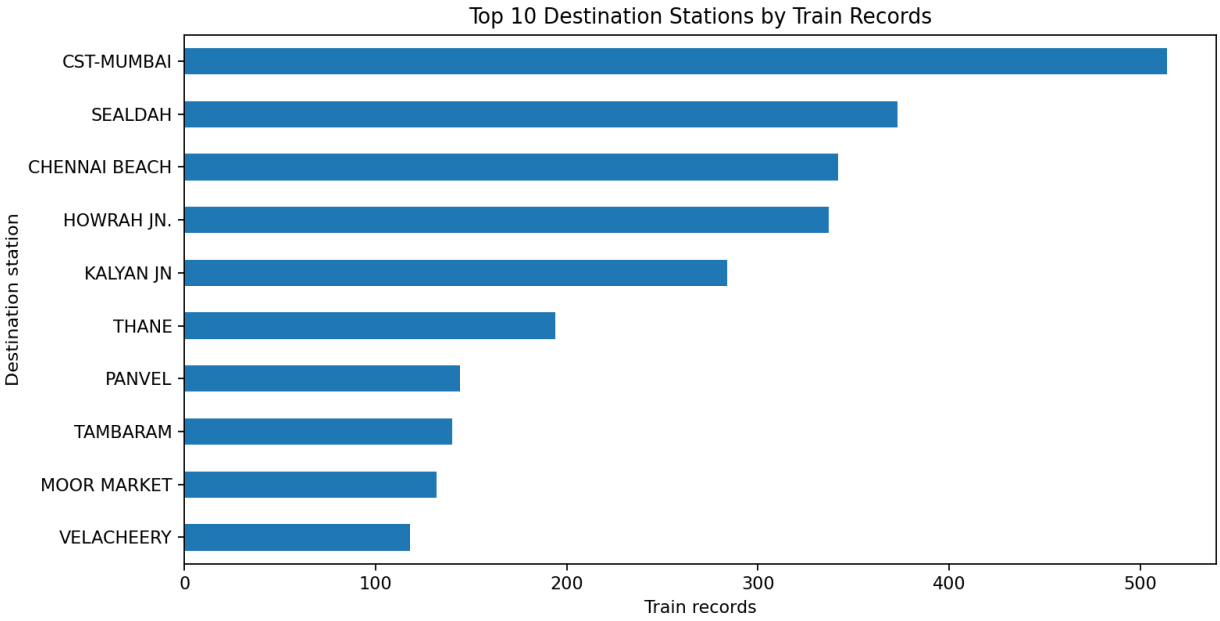
Train records by operating day



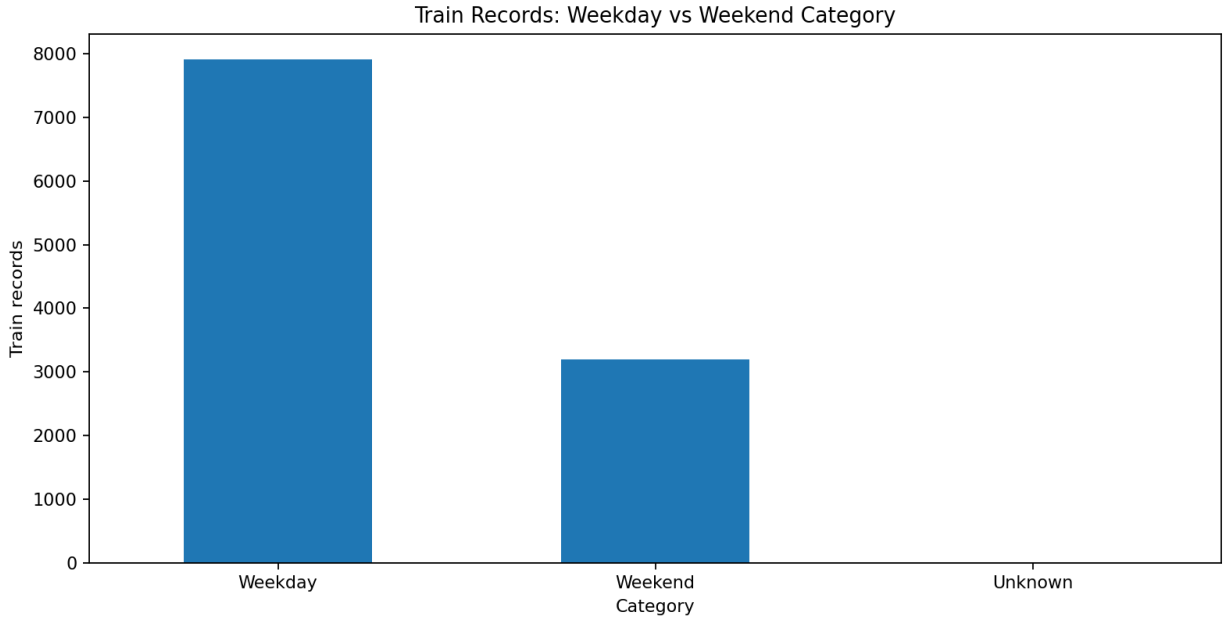
Top source stations



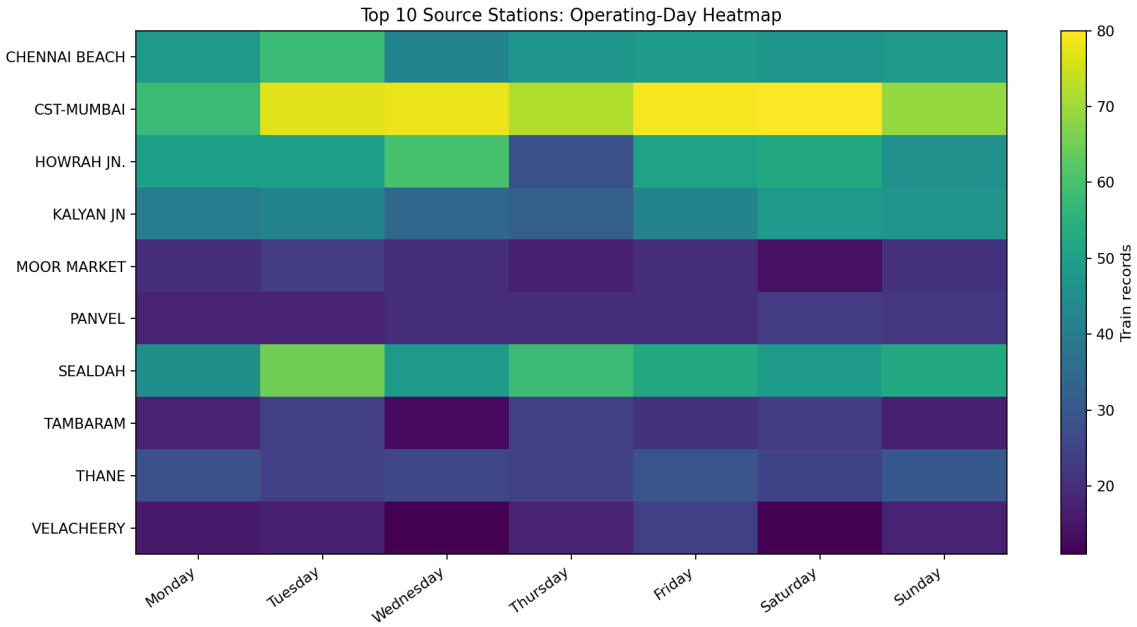
Top destination stations



Weekday vs weekend



Source station/day heatmap



4. Key Findings

• The dataset contains 11,113 rows and 5 columns before cleaning.

• There are 921 unique source stations and 924 unique destination stations.

• CST-MUMBAI is the most frequently recorded source station (513 records); CST-MUMBAI is the most frequently recorded destination station (514 records).

• Friday has the highest normalized day count (1,649); Monday has the lowest (1,503).

• Normalized day categories contain 7,918 weekday records and 3,195 weekend records.

• The raw day field includes 1,153 records whose value ends in 'd'. This suffix was preserved in days_raw and flagged separately; its exact business meaning should be confirmed with the dataset owner.

• The average-trains-per-day column is calculated as each source station's row count divided by seven. It is a simple normalized average, not a timetable-level estimate, because the source data is a list of train records rather than daily service counts for every calendar date.

5. Recommendations

• Confirm the meaning of the trailing 'd' in day values before using it for operational decisions; the raw value is retained in days_raw and Daily_Flag_Suffix_d.

• Use the cleaned Operating_Day column for day-wise charts and filters so values such as 'Saturday' and 'Saturdayd' are grouped under Saturday while preserving the original value.

• Review high-volume source and destination stations for staffing, platform planning, and timetable validation; treat record counts as dataset frequency rather than real-time passenger demand.

• For a stronger day-of-week correlation analysis, obtain dated service records or an explicit one-hot operating calendar. A single categorical operating-day field supports distribution comparisons, but not a meaningful causal correlation claim by itself.

Tools: Python, pandas, NumPy, Matplotlib, openpyxl. See the accompanying Excel workbook and source code for detailed tables and reproducibility.